

Workshop: From Descriptive Statistics to Causal Thinking in Smart City Road Safety

Summer School 2026
Causal Inference and Data Science for Smart Cities

Context

Smart cities generate large amounts of data about mobility, infrastructure, road conditions, traffic, and citizen safety. However, working with urban data requires more than calculating averages or producing visualizations. A key challenge is to move from a purely descriptive question, such as *where are accidents more frequent?*, to a causal question, such as *would improving road lighting reduce the probability of accidents?*

In this workshop, students will work with the dataset:

`smart_city_road_safety_.csv`

Instead, students must decide which tables, summaries, and plots are useful to answer a set of descriptive and causal questions. The workshop is designed to encourage students to think critically about what the data can show, what it cannot show, and how causal reasoning can improve the interpretation of urban road safety problems.

General Objective

To guide students from a basic descriptive analysis of a smart city road safety dataset toward a causal interpretation of road accident risk using variables, graphs, and causal questions.

Specific Objectives

By the end of the workshop, students should be able to:

- Identify the main variables in a road safety dataset.
- Produce simple descriptive statistics and visual summaries.
- Compare accident patterns across road, traffic, driver, and environmental conditions.
- Distinguish descriptive questions from causal questions.
- Identify possible outcome variables, treatment variables, confounders, mediators, and bad controls.

- Use graphs to suggest possible causal mechanisms.
- Explain why association is not the same as causation.

1 Dataset Description

The dataset contains observations related to road safety events in a smart city environment. Each row represents a traffic situation or road safety observation. The variables include characteristics of the environment, the road, the driver, the traffic context, and the accident outcome.

Main Variables

Variable	Interpretation
Weather	Weather conditions during the observation.
Road_Type	Type of road where the observation occurred.
Time_of_Day	Time period of the day.
Traffic_Density	Level of traffic density.
Speed_Limit	Speed limit in the road segment.
Number_of_Vehicles	Number of vehicles involved or present in the traffic situation.
Driver_Alcohol	Indicator of alcohol involvement by the driver.
Accident_Severity	Severity level of the accident.
Road_Condition	Physical or surface condition of the road.
Vehicle_Type	Type of vehicle involved.
Driver_Age	Age of the driver.
Driver_Experience	Driving experience of the driver.
Road_Light_Condition	Lighting condition of the road.
Accident	Indicator of whether an accident occurred.

Important distinction: `Accident` can be treated as the main outcome variable because it indicates whether an accident occurred. `Accident_Severity` should be interpreted carefully, because severity is only meaningful once an accident has occurred.

2 Part I: First Exploration of the Dataset

Purpose

The first part of the workshop focuses on understanding the structure of the dataset before making any causal claims.

Guiding Questions

Students should begin by answering the following questions:

1. How many observations are in the dataset?
2. Which variables are numerical and which are categorical?

3. Are there missing values?
4. What are the most frequent categories for each categorical variable?
5. What are the minimum, maximum, mean, median, and standard deviation of the numerical variables?
6. What proportion of observations correspond to accident cases?
7. How frequent are the different levels of accident severity?

Expected Outputs

Students should produce:

- A table with the type of each variable.
- A table of descriptive statistics for numerical variables.
- Frequency tables for categorical variables.
- A short paragraph describing the general structure of the dataset.

Suggested Graphs

Students may use:

- Bar charts for categorical variables.
- Histograms or density plots for numerical variables.
- Boxplots for driver age, driver experience, speed limit, or number of vehicles.
- A pie chart or bar chart to show the proportion of accident and non-accident cases.

3 Part II: Descriptive Road Safety Questions

Purpose

This section asks students to use descriptive statistics and visualizations to understand accident patterns. At this stage, the objective is not to make causal claims, but to identify possible associations.

Descriptive thinking asks: What patterns do we observe in the data?

3.1 Question 1: When do accidents occur more frequently?

Students should explore the relationship between accidents and `Time_of_Day`.

Questions to Answer

1. Are accidents more frequent during specific periods of the day?
2. Is accident severity different across time periods?
3. Does the distribution of traffic density change across time periods?

Suggested Graphs

- Bar chart of accident rate by `Time_of_Day`.
- Stacked bar chart of `Accident_Severity` by `Time_of_Day`.
- Boxplot or bar chart of `Traffic_Density` by `Time_of_Day`.

3.2 Question 2: Are some road conditions associated with more accidents?

Students should explore the relationship between accidents and `Road_Condition`.

Questions to Answer

1. Which road conditions have the highest accident rate?
2. Are poor road conditions associated with more severe accidents?
3. Are road conditions related to weather conditions?

Suggested Graphs

- Bar chart of accident rate by `Road_Condition`.
- Stacked bar chart of accident severity by `Road_Condition`.
- Heatmap or grouped bar chart comparing `Weather` and `Road_Condition`.

3.3 Question 3: Does road lighting seem related to accident risk?

Students should explore `Road_Light_Condition` and `Accident`.

Questions to Answer

1. Are accidents more frequent under poor lighting conditions?
2. Does accident severity vary by lighting condition?
3. Are poor lighting conditions more common at specific times of day or on specific road types?

Suggested Graphs

- Bar chart of accident rate by `Road_Light_Condition`.
- Stacked bar chart of `Accident_Severity` by `Road_Light_Condition`.
- Grouped bar chart of `Road_Light_Condition` by `Time_of_Day`.
- Grouped bar chart of `Road_Light_Condition` by `Road_Type`.

3.4 Question 4: Are driver characteristics associated with accident risk?

Students should explore `Driver_Age`, `Driver_Experience`, and `Driver_Alcohol`.

Questions to Answer

1. Do accident cases involve younger or less experienced drivers?
2. Is alcohol involvement associated with a higher accident rate?
3. Is alcohol involvement associated with more severe accidents?
4. Does driver experience change the relationship between alcohol involvement and accidents?

Suggested Graphs

- Boxplot of `Driver_Age` by `Accident`.
- Boxplot of `Driver_Experience` by `Accident`.
- Bar chart of accident rate by `Driver_Alcohol`.
- Grouped bar chart of accident severity by `Driver_Alcohol`.

3.5 Question 5: How do traffic variables relate to accident risk?

Students should explore `Traffic_Density`, `Speed_Limit`, and `Number_of_Vehicles`.

Questions to Answer

1. Are accident rates higher when traffic density is higher?
2. Are speed limits related to accident occurrence?
3. Do accident cases involve more vehicles on average?
4. Does the relationship between traffic density and accidents vary by road type?

Suggested Graphs

- Bar chart of accident rate by `Traffic_Density`.
- Boxplot of `Speed_Limit` by `Accident`.
- Boxplot of `Number_of_Vehicles` by `Accident`.
- Grouped bar chart of accident rate by `Traffic_Density` and `Road_Type`.

4 Part III: From Descriptive Questions to Causal Questions

Purpose

After identifying patterns, students must now ask whether some of these patterns could be interpreted causally. This requires moving from observed associations to possible interventions.

Causal thinking asks: What would happen to accident risk if we changed one condition while keeping other relevant factors comparable?

4.1 Statistical Association vs. Causal Effect

A descriptive question may be written as:

$$P(\text{Accident} = 1 \mid \text{Road_Light_Condition})$$

This asks whether accident probability differs across observed lighting conditions.

A causal question may be written as:

$$P(\text{Accident} = 1 \mid do(\text{Road_Light_Condition}))$$

This asks what would happen to accident probability if the road lighting condition were changed by an intervention.

The difference is fundamental. The first expression describes an observed association. The second expression represents an intervention.